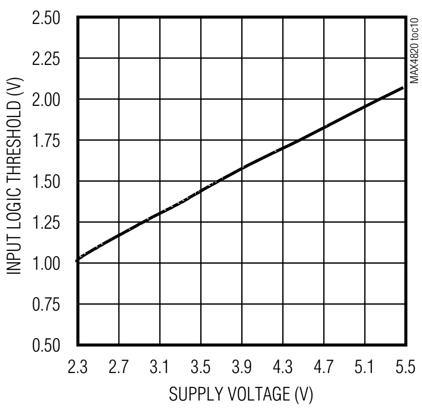
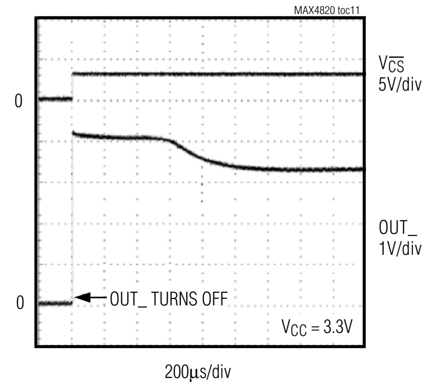
Typical Operating Characteristics (continued)

($V_{COM} = V_{CC}$, T_A = -40°C to +85°C, unless otherwise noted. Typical values are at T_A = +25°C.)





Pin Description

PIN				NAME	FUNCTION
MAX4820		MAX4821			
THIN QFN	TSSOP	THIN QFN	TSSOP		
1	3	1	3	$\overline{\text{RESET}}$	Reset Input. Drive $\overline{\text{RESET}}$ low to clear all latches and registers (all outputs are turned off). $\overline{\text{RESET}}$ overrides all other inputs. If $\overline{\text{RESET}}$ and $\overline{\text{SET}}$ are pulled low at the same time, then $\overline{\text{RESET}}$ takes precedence.
2	4	2	4	$\overline{\text{CS}}$	Chip-Select Input. **MAX4820:** Drive $\overline{\text{CS}}$ low to select the device. When $\overline{\text{CS}}$ is low, data at DIN is clocked into the 8-bit shift register on SCLK's rising edge. Drive $\overline{\text{CS}}$ from low to high to latch the data to the registers and activate the appropriate relays. **MAX4821:** Drive $\overline{\text{CS}}$ low to select the device and set level on LVL. Drive $\overline{\text{CS}}$ from low to high to latch the address and level data to the output.
3	5	—	—	DIN	Serial-Data Input
4	6	—	—	SCLK	Serial-Clock Input
5	7	—	—	DOUT	Serial-Data Output. DOUT is the output of the 8-bit shift register. This output can be used to daisy chain multiple MAX4820s. The data at DOUT appears synchronous to SCLK's falling edge.
6	8	—	—	N.C.	No Connection
7	9	7	9	GND	Ground
8	10	8	10	OUT8	Open-Drain Output 8. Connect OUT8 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
9	11	9	11	OUT7	Open-Drain Output 7. Connect OUT7 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
10, 16	12, 18	10, 16	12, 18	PGND	Power Ground. PGND is a return for the output sinks. Connect PGND pins together and to GND.
11	13	11	13	OUT6	Open-Drain Output 6. Connect OUT6 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.

+3.3V/+5V, 8-Channel, Cascadable Relay Drivers with Serial/Parallel Interface

Pin Description (continued)

PIN				NAME	FUNCTION
MAX4820		MAX4821			
THIN QFN	TSSOP	THIN QFN	TSSOP		
12	14	12	14	OUT5	Open-Drain Output 5. Connect OUT5 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
13	15	13	15	COM	Common Free-Wheeling Diodes. Connect COM to V_{CC}. COM can also be connected to a separate supply that is higher than V_{CC}. In that case, bypass V_{CC} to GND with a 0.1µF capacitor.
14	16	14	16	OUT4	Open-Drain Output 4. Connect OUT4 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
15	17	15	17	OUT3	Open-Drain Output 3. Connect OUT3 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
17	19	17	19	OUT2	Open-Drain Output 2. Connect OUT2 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
18	20	18	20	OUT1	Open-Drain Output 1. Connect OUT1 to the low side of a relay coil. This output is pulled to PGND when activated, but otherwise is high impedance.
19	1	19	1	V_{CC}	Input Supply Voltage. Bypass V_{CC} to GND with a 0.1µF capacitor.
20	2	20	2	$\overline{SET}$	Set Input. Drive $\overline{SET}$ low to set all latches and registers high (all outputs are turned on). $\overline{SET}$ overrides all parallel and serial control inputs. $\overline{RESET}$ overrides $\overline{SET}$ under all conditions.
—	—	3	5	LVL	Level Input. LVL determines whether the selected address is switched on or off. A logic high on LVL switches on the addressed output. A logic low on LVL switches off the addressed output.
—	—	4	6	A0	Digital Address “0” Input. (See Table 2 for address mapping.)
—	—	5	7	A1	Digital Address “1” Input. (See Table 2 for address mapping.)
—	—	6	8	A2	Digital Address “2” Input. (See Table 2 for address mapping.)
—	—	—	—	EP	Exposed Pad. Connect exposed pad to GND.

Detailed Description

The MAX4820/MAX4821 8-channel relay drivers offer built-in kickback protection and drive +3.3V/+5V non-latching or dual-coil-latching relays. These devices are especially useful when driving +3V relays. Each independent open-drain output features a 2Ω on-resistance and is guaranteed to sink 70mA (min) load current. Both devices consume less than 50µA (max) quiescent current and feature 1µA (min) output off-leakage current.

The MAX4820 features an SPI/QSPI/MICROWIRE-compatible serial interface. Input data is shifted into an 8-bit shift register and latched to the outputs when $\overline{CS}$ transitions from low to high. Each data bit in the shift register corresponds to a specific output, allowing independent control of all outputs.

The MAX4821 features a 4-bit (A0, A1, A2, LVL) parallel input interface. The three bits (A0, A1, A2) determine the output address, and LVL determines whether the selected output is switched on or off. Data is latched to the outputs when $\overline{CS}$ transitions from low to high.

Both devices feature separate set and reset functions that allow the user to turn on or turn off all outputs simultaneously with a single control line. Built-in hysteresis (Schmidt trigger) on all digital inputs allows this device to be used with slow rising and falling signals, such as those from optocouplers or RC power-up initialization circuits. The MAX4820/MAX4821 are available in 20-pin TSSOP and space-saving 20-pin Thin QFN packages.